

Preface

Nothing puzzles me more than time and space; and yet nothing troubles me less . . .

These words, by the English essayist Charles Lamb in his 1810 letter to Thomas Manning, provide a concise summary of this book. Of course, he was not thinking of Statistics, nor Science, nor Statistical Science, rather the more ephemeral notion of space and time. But here, in the physical world, there are scientific questions to resolve and predictions to make, and understanding the effects of dependencies across time and space is a crucial part. Those dependencies are in there somewhere, and we too are puzzled by them.

Up until the mid-twentieth century, Statistics' response to this puzzle was often to ignore the complicated structure, or to find clever ways to remove it. Interestingly, Charles Lamb seemed to be in tune with this, as he finished his sentence by saying, "as I never think about them." As Statistics has progressed through the twentieth century and moved into the twenty-first, spatial and temporal statistical methodology has been incorporated more and more into the scientific models of our world and indeed of our universe. As technology has improved, Statistics has been given a Rosetta Stone to begin to unlock Science's mysteries, from the molecular to the global to the cosmological.

In the beginning, there were data. Then, there were theories, formed from data, and those theories rose and fell according their agreement with new data. Data are central, and they should be cared for accordingly. To do good Science, databases should be fully documented and algorithms ("black boxes") involved in creating them should be in the public domain.

However, data alone do not tell us all that much about our world. When we look at data, how do we know if what we are seeing is "signal" as opposed to "noise"? How do we compare two sources of data—what is the basis of a comparison?

Similarly, theories (or models) by themselves are not often the best descriptions of the real world. What can be said about processes on scales at which

there were no observations? What about uncertainties in parameters, or forcings, or interactions with other processes? Indeed, the key is the proper blending of such models and the data. Sometimes this might be done informally, for example, by taking a simulated field, simulated from a mathematical (say) model, and visually comparing it to a field of actual data. From the visual comparison, a deficiency in the model might be obvious, which might lead to a parameter adjustment, or even a new parameter. Then a *new* simulation might be implemented and a new visual comparison made, and so forth. This is one way to combine data and model, but it is not a very efficient way to deal with either. Indeed, the power of Statistical Science is that it provides several frameworks in which to combine data and model, in optimal ways, for the purpose of scientific inference. It might seem strange that there is not just one framework in which to carry out inferences, but even Statistics has its tribes. However, the element that is common to all is an attempt to partition variability and to quantify uncertainty.

In this book, we take the firm stand that the best paradigm (to date) in which to partition variability and quantify uncertainty is the *hierarchical statistical model*. Such a model explicitly acknowledges uncertainty in the data, different from that in the process and parameters, and then it accommodates the uncertainty in the process (and finally in parameters, if necessary). We have used color in key places in our exposition, to distinguish between the parts of the hierarchical model concerned with data (green), with processes (blue) and with parameters (purple).

Hierarchical thinking (i.e., hierarchical modeling) is intimately tied to conditional thinking (i.e., modeling with conditional probabilities). Indeed, it is our perspective that conditional thinking is the aforementioned Rosetta Stone: It allows us to separately partition the effects of measurement error and scales of variability below the resolution of our data, conditioned on the process at possibly some other scale. Similarly, conditional thinking allows us to model a spatio-temporal process as it actually *evolves* through time, as opposed to just accounting for its marginal dependencies. Equally important, it allows us to use spatio-temporal dependencies in errors and/or parameters as a proxy for unknown and unknowable processes. Its impact goes even deeper, in that it allows parameters themselves to be dependent on other processes or other sources of data. Finally, it makes clear that if some components of variability are not interesting for a particular question, then the end result should *not* look like the data. Any unwanted components (e.g., measurement error) are filtered out: What you see (data) is not always what you want to get (process).

Critically, in the presence of data, conditional thinking allows sensible trade-offs to be made between data availability, process complexity, and computational complexity. Combined with the ever-expanding computational tools of the twenty-first century, the hierarchical statistical framework provides the structure to tackle important questions in Energy, Climate, Environment, Food, Finance, and so on. This will be a century of massive (spatio-temporal) datasets

collected to answer Society's dominant questions. In this book, we are particularly interested in inference for Climate and the Environment, where processes at small spatio-temporal scales influence those at larger scales, and vice versa. The questions to be resolved are fundamental to sustaining our planet, they involve complex spatio-temporal phenomena, *and* they are inherently statistical.

Our lives are spent marching through a space–time continuum, but space is different from time. We can (and often do) visit the same place over and over but always at different, ordered time points. We can go north, south, east, and west, up and down, but only ever forward from the past to the present and into the future. Any study of a spatio-temporal phenomenon needs to respect this difference. In this book, we have proposed spatio-temporal statistical methodologies that align with the underlying science, and we have found that hierarchical thinking is a natural way to achieve this alignment.

In the pages that follow, we have deliberately tried to build a bridge between the twentieth and twenty-first centuries in our presentation of spatio-temporal statistics. There are strong and powerful traditions that have developed in the last few decades and, even if they are not hierarchical, they provide the tools and motivation that can be used in hierarchical thinking. In some cases, hierarchical approaches that may be appealing in principle may be out of our reach in terms of timely implementation.

The official ftp site associated with the book can be found at:

ftp://ftp.wiley.com/public/sci_tech_med/spatio_temporal_data

In addition to posting errata, we will post supplemental material that we hope will be helpful to the reader.

As mentioned above, we have questions to resolve and predictions to make, so let's get started . . .

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